WORKED SOLUTION

STEP I 2008, Question 7

Pure • Rotations and Distances

STEP I 2008, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q7

The key idea

A perpendicular distance to a line through the origin becomes easy once we rotate the plane so that the line lies along the x-axis: the distance is then just the size of the new y-coordinate. The rotation formula from the first part does exactly this for two of the three sides. For the final part, the trick is to drop the modulus signs: the three *signed* distances always add up to the same constant, and the moduli can only make the total bigger.

Solution

The rotation formulae

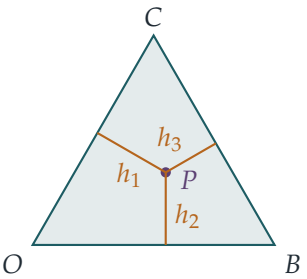
Write $x = r\cos\theta$, $y = r\sin\theta$, where $r = OP$ and θ is the angle from the positive x-axis to OP. Rotating OP clockwise by 60° keeps r and changes the angle to $\theta - 60^\circ$, so the new y-coordinate is

$$\begin{aligned} r\sin(\theta - 60^\circ) &= r\sin\theta\cos 60^\circ - r\cos\theta\sin 60^\circ \\ &= \tfrac{1}{2}y - \tfrac{\sqrt{3}}{2}x = \tfrac{1}{2}(y - \sqrt{3}\,x), \end{aligned}$$

as required. For an anticlockwise rotation the angle becomes $\theta + 60^\circ$, and the same expansion with the sign of $\sin 60^\circ$ reversed gives the new y-coordinate

$$\tfrac{1}{2}(y + \sqrt{3}\,x).$$

The three distances



h_1. The line OC makes an angle of 60° with the x-axis. Rotating the whole picture clockwise by 60° about O carries OC onto the x-axis and preserves distances. In the rotated picture the distance from P to the x-axis is the modulus of its new y-coordinate, so

$$h_1 = \tfrac{1}{2}|\, y - \sqrt{3}\, x\,|.$$

h_2. The line OB is the x-axis, so $h_2 = |y|$.

h_3. Now rotate anticlockwise by 60° about O. This sends $B = (1,0)$ to $(\tfrac{1}{2}, \tfrac{1}{2}\sqrt{3})$ and C (at angle 60°) to the point at angle 120°, namely $(-\tfrac{1}{2}, \tfrac{1}{2}\sqrt{3})$. So the line BC becomes the horizontal line $y = \tfrac{1}{2}\sqrt{3}$, while P acquires the y-coordinate $\tfrac{1}{2}(y + \sqrt{3}\, x)$. Hence

$$h_3 = \left|\tfrac{1}{2}(y + \sqrt{3}\, x) - \tfrac{1}{2}\sqrt{3}\right| = \tfrac{1}{2}|\, \sqrt{3}\, x + y - \sqrt{3}\,|.$$

(Equivalently: BC has equation $\sqrt{3}\, x + y = \sqrt{3}$, and the usual distance formula gives the same answer, since $\sqrt{3+1} = 2$.)

The final part

Define the signed quantities

$$s_1 = \tfrac{1}{2}(\sqrt{3}\, x - y), \qquad s_2 = y, \qquad s_3 = \tfrac{1}{2}(\sqrt{3} - \sqrt{3}\, x - y),$$

so that $h_i = |s_i|$ for $i = 1, 2, 3$. Adding,

$$s_1 + s_2 + s_3 = \tfrac{1}{2}\sqrt{3}\, x - \tfrac{1}{2}y + y + \tfrac{1}{2}\sqrt{3} - \tfrac{1}{2}\sqrt{3}\, x - \tfrac{1}{2}y = \tfrac{1}{2}\sqrt{3}$$

for *every* point (x, y).

Ask yourself How do the signs of s_1, s_2, s_3 relate to the triangle?

Each s_i is a linear expression that vanishes on one side of the triangle. Testing the centroid $(\tfrac{1}{2}, \tfrac{1}{6}\sqrt{3})$, which is inside, gives $s_1 = s_2 = s_3 = \tfrac{1}{6}\sqrt{3} > 0$. So $s_2 \geq 0$ is the side of OB containing C, $s_1 \geq 0$ is the side of OC containing B, and $s_3 \geq 0$ is the side of BC containing O. The closed triangle OBC is exactly the intersection of these three half-planes, so

$$P \text{ lies on or in } OBC \iff s_1 \geq 0,\ s_2 \geq 0,\ s_3 \geq 0.$$

Now, for any real number s we have $|s| \geq s$, with equality if and only if $s \geq 0$. Therefore

$$h_1 + h_2 + h_3 = |s_1| + |s_2| + |s_3| \;\geq\; s_1 + s_2 + s_3 = \tfrac{1}{2}\sqrt{3},$$

with equality if and only if $|s_i| = s_i$ for each i, that is, if and only if all three s_i are non-negative. By the previous paragraph, this happens exactly when P lies on or in the triangle OBC. This proves both directions at once:

$$h_1 + h_2 + h_3 = \tfrac{1}{2}\sqrt{3} \iff P \text{ lies on or in triangle } OBC.$$

Note also that outside the triangle the sum is strictly greater than $\frac{1}{2}\sqrt{3}$.

Checks

- At a vertex, say $P = B = (1, 0)$: $h_1 = \frac{1}{2}\sqrt{3}$, $h_2 = 0$, $h_3 = 0$, sum $\frac{1}{2}\sqrt{3}$. Good.
- At the centroid each distance is $\frac{1}{6}\sqrt{3}$, and three of them make $\frac{1}{2}\sqrt{3} \approx 0.8660$, the height of the triangle.
- Outside, at $P = (2, 0)$: $h_1 = \sqrt{3}$, $h_2 = 0$, $h_3 = \frac{1}{2}\sqrt{3}$, sum $\frac{3}{2}\sqrt{3} > \frac{1}{2}\sqrt{3}$.
- We compared all three formulae with directly computed point-to-line distances at 10^5 random points (agreement to 10^{-15}), and tested the "if and only if" at 2×10^5 points, including many on the edges, with no exceptions.

Common slips

- Getting the sign of the rotation wrong: clockwise means the angle *decreases*, giving $\theta - 60^\circ$.
- Proving only one direction. Checking that the sum is $\frac{1}{2}\sqrt{3}$ inside the triangle is not enough; you must also show it is *not* $\frac{1}{2}\sqrt{3}$ outside. The inequality $|s| \geq s$ handles both at once.
- Forgetting the modulus signs in h_1, h_2, h_3: distances are never negative.

Beyond STEP

This is Viviani's theorem: in an equilateral triangle the sum of the distances from an interior point to the sides equals the height. The signed distances s_1, s_2, s_3 are (up to scaling) the *barycentric* or *trilinear coordinates* of P, which always sum to a constant and are all non-negative exactly inside the triangle; they are a standard tool in geometry, computer graphics and convex analysis. The rotation formula is the second row of a 2×2 rotation matrix, and Strang, *Introduction to Linear Algebra*, is a good place to see rotations as linear maps.

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